

Supplemental Material: A Neutral Network Parametrized Collision Rate Model for < 3 nm Titanium Dioxide Nanoclusters

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1. MD simulation condition

Table S1. MD simulation conditions. D_p is the nanocluster diameter, N is the number of atoms in a nanocluster, b is the impact parameter, v_0 is the initial velocity, db and dv_0 are the increments of b and v_0 , respectively.

$D_{p,i}$ [nm]	$D_{p,j}$ [nm]	N_i	N_j	b [nm]	v_0 [m/s]	db [nm]	dv_0 [m/s]
0.6	0.6	9	9	0-7.8	20-200	0.2	20
0.6	0.6	9	9	0-7.8	250-700	0.2	50
0.6	0.8	9	24	0-7.8	20-200	0.2	20
0.6	0.8	9	24	0-7.8	250-700	0.2	50
0.6	1.2	9	84	0-7.8	20-200	0.2	20
0.6	1.2	9	84	0-7.8	250-700	0.2	50
0.6	1.4	9	132	0-7.8	20-200	0.2	20
0.6	1.4	9	132	0-7.8	250-700	0.2	50
0.6	1.6	9	198	0-7.8	20-200	0.2	20
0.6	1.6	9	198	0-7.8	250-700	0.2	50
0.6	1.8	9	291	0-7.8	20-200	0.2	20
0.6	1.8	9	291	0-7.8	250-700	0.2	50
0.6	2.0	9	390	0-7.8	20-200	0.2	20
0.6	2.0	9	390	0-7.8	250-700	0.2	50
0.6	2.2	9	534	1.4-7.8	20-200	0.2	20
0.6	2.2	9	534	1.5-7.8	250-700	0.2	50
0.6	2.4	9	696	1.6-7.8	20-200	0.2	20
0.6	2.4	9	696	0-7.8	250-700	0.2	50
0.6	2.6	9	882	0-7.8	20-200	0.2	20
0.6	2.6	9	882	0-7.8	250-700	0.2	50
0.6	2.8	9	1,098	0-7.8	20-200	0.2	20
0.6	2.8	9	1,098	0-7.8	250-700	0.2	50
0.6	3.0	9	1,356	1.8-7.8	20-200	0.2	20
0.6	3.0	9	1,356	1.8-7.8	250-700	0.2	50
1.0	0.6	9	48	0-7.8	20-200	0.2	20
1.0	0.6	9	48	0-4.8	250-700	0.2	50
1.0	0.8	48	24	0-7.8	20-200	0.2	20
1.0	0.8	48	24	0-4.8	250-700	0.2	50
1.0	1.0	48	48	0-4.8	20-200	0.2	20
1.0	1.0	48	48	0-4.8	250-700	0.2	50
1.0	1.2	48	84	0-4.8	20-200	0.2	20
1.0	1.2	48	84	0-4.8	250-700	0.2	50
1.0	1.4	48	132	0-4.8	20-200	0.2	20
1.0	1.4	48	132	0-4.8	250-700	0.2	50
1.0	1.6	48	198	0-4.8	20-200	0.2	20
1.0	1.6	48	198	0-4.8	250-700	0.2	50
1.0	1.8	48	291	0-4.8	20-200	0.2	20
1.0	1.8	48	291	0-4.8	250-700	0.2	50
1.0	2.0	48	390	1.0-3.8	20-200	0.2	20
1.0	2.0	48	390	0-6.8	250-700	0.2	50

1.0	2.2	48	534	0-3.8	20-200	0.2	20
1.0	2.2	48	534	0-6.8	250-700	0.2	50
1.0	2.4	48	696	0-3.8	20-200	0.2	20
1.0	2.4	48	696	0-6.8	250-700	0.2	50
1.0	2.6	48	882	0-3.8	20-200	0.2	20
1.0	2.6	48	882	0-6.8	250-700	0.2	50
1.0	2.8	48	1,098	0-3.8	20-200	0.2	20
1.0	2.8	48	1,098	0-6.8	250-700	0.2	50
1.0	3.0	48	1,356	0-7.8	20-200	0.2	20
1.0	3.0	48	1,356	0-2.8	250-700	0.2	50
2.0	0.8	390	24	0-7.8	20-200	0.2	20
2.0	0.8	390	24	0-4.8	250-700	0.2	50
2.0	1.2	390	84	1.6-4.8	20-200	0.2	20
2.0	1.2	390	84	1.6-6.8	250-700	0.2	50
2.0	1.4	390	132	1.6-4.8	20-200	0.2	20
2.0	1.4	390	132	1.6-6.8	250-700	0.2	50
2.0	1.6	390	198	1.6-4.8	20-200	0.2	20
2.0	1.6	390	198	1.6-6.8	250-700	0.2	50
2.0	1.8	390	291	1.6-4.8	20-200	0.2	20
2.0	1.8	390	291	1.6-6.8	250-700	0.2	50
2.0	2.0	390	390	1.6-4.8	20-200	0.2	20
2.0	2.0	390	390	0-6.8	250-700	0.2	50
2.0	2.2	390	534	1.6-4.8	20-200	0.2	20
2.0	2.2	390	534	1.6-6.8	250-700	0.2	50
2.0	2.4	390	696	2.2-4.8	20-200	0.2	20
2.0	2.4	390	696	2.2-7.8	250-700	0.2	50
2.0	2.6	390	882	2.0-3.8	20-200	0.2	20
2.0	2.6	390	882	2.0-7.8	250-700	0.2	50
2.0	2.8	390	1,098	2.2-3.8	20-200	0.2	20
2.0	2.8	390	1,098	2.2-7.8	250-700	0.2	50
2.0	3.0	390	1,356	2.2-3.8	20-200	0.2	20
2.0	3.0	390	1,356	2.2-7.8	250-700	0.2	50
3.0	3.0	1,356	1,356	3.0-3.8	20-200	0.2	20
3.0	3.0	1,356	1,356	3.0-7.8	250-700	0.2	50

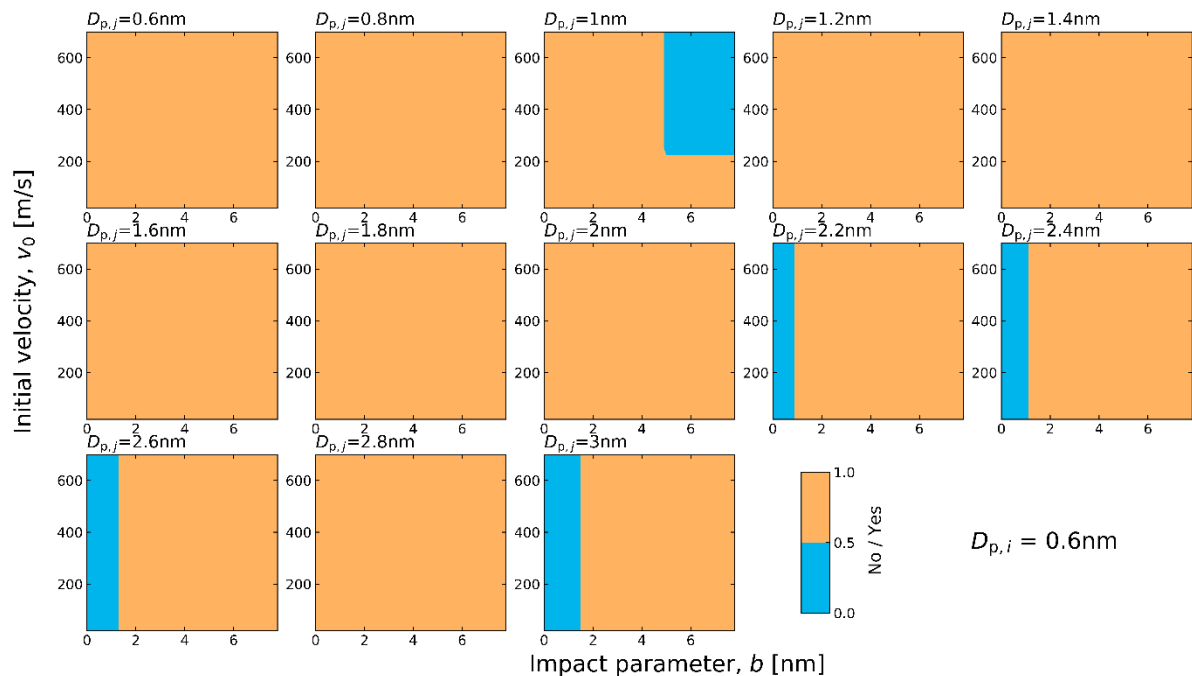


Figure S1. Calculation range map for $D_{p,i} = 0.6$ nm. The orange region is calculated while blue is skipped since it is obviously collision/no-collision. The calculation of $(D_{p,i}, D_{p,j}) = (0.6$ nm, 1.0 nm) was not performed but it referred from $(D_{p,i}, D_{p,j}) = (0.6$ nm, 1.0 nm).

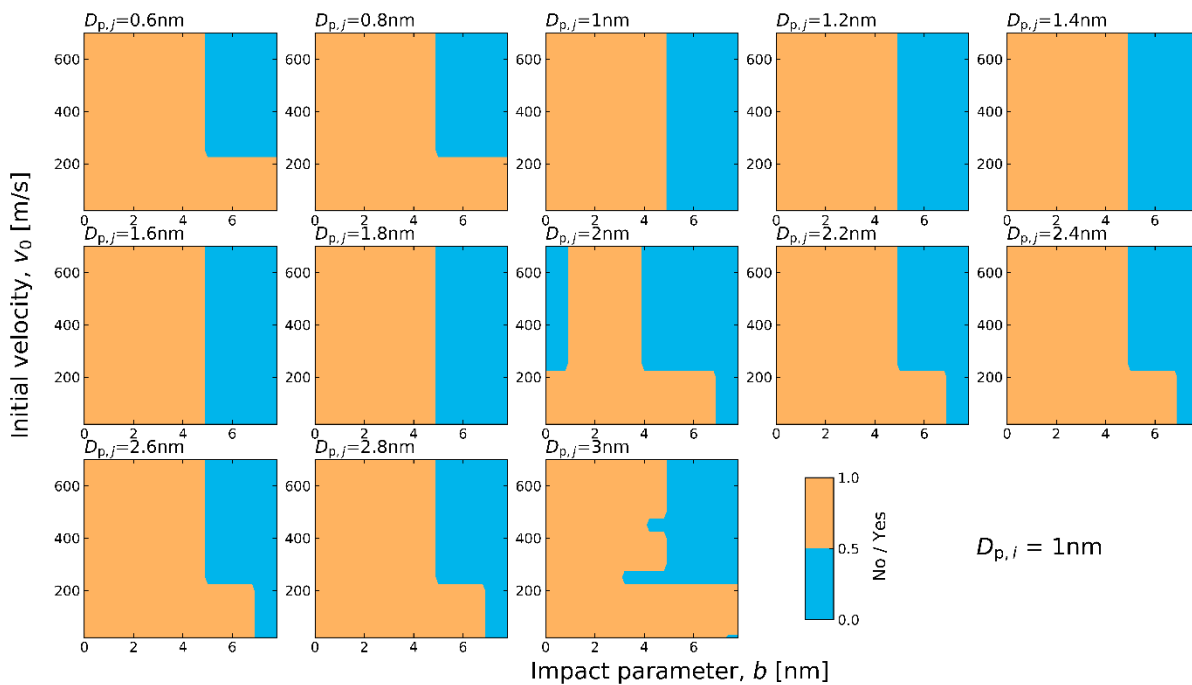


Figure S2. Calculation range map for $D_{p,i} = 1.0$ nm. The orange region is calculated while blue is skipped since it is obviously collision/no-collision.

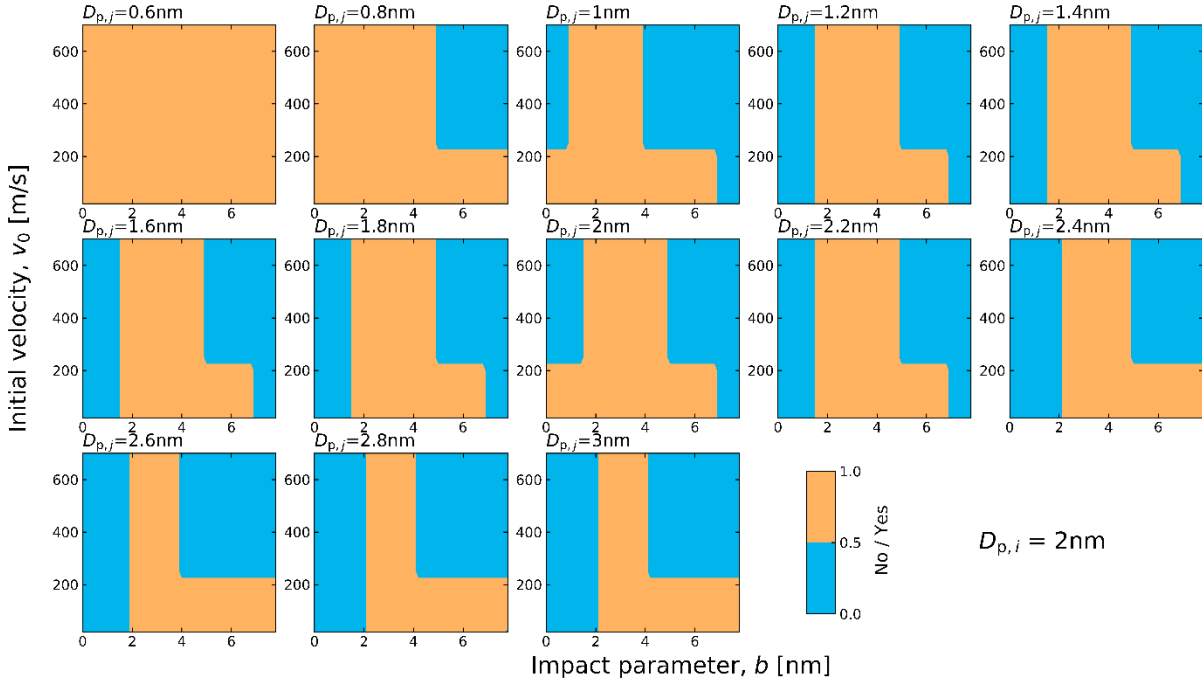


Figure S3. Calculation range map for $D_{p,i} = 2.0$ nm. The orange region is calculated while blue is skipped since it is obviously collision/no-collision. The calculations of $(D_{p,i}, D_{p,j}) = (2.0$ nm, 0.6 nm) and $(D_{p,i}, D_{p,j}) = (2.0$ nm, 1.0 nm) were not performed but these were referred from $(D_{p,i}, D_{p,j}) = (0.6$ nm, 2.0 nm) and $(D_{p,i}, D_{p,j}) = (1.0$ nm, 2.0 nm), respectively.

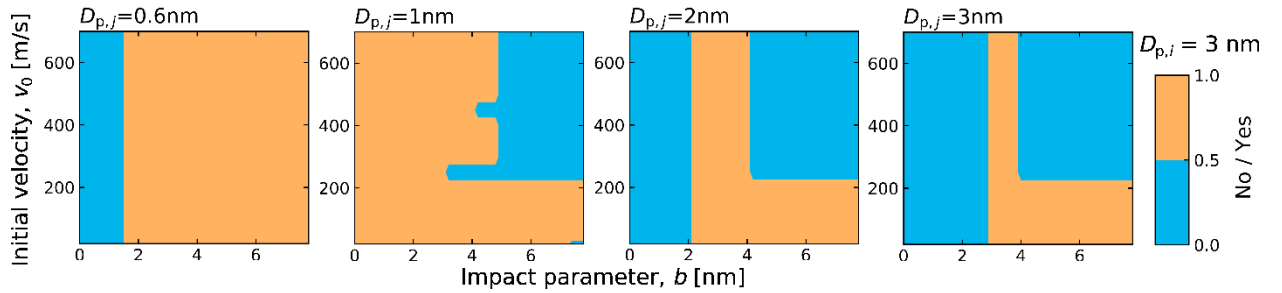


Figure S4. Calculation range map for $D_{p,i} = 3.0$ nm. The orange region is calculated while blue is skipped since it is obviously collision/no-collision.

2. Matsui-Akaogi potential parameters

Table S2. Potential parameters used in the Matsui-Akaogi potential for each atom type combinations

	A_{kl} [kcal mol ⁻¹]	ρ_{kl} [Å ⁻¹]	C_{kl} [Å ⁶ kcal mol ⁻¹]
Ti-Ti	7.18×10^5	6.49	121.0
O-O	2.72×10^5	4.27	290.4
Ti-O	3.91×10^5	5.15	696.9

3. Probability heat maps

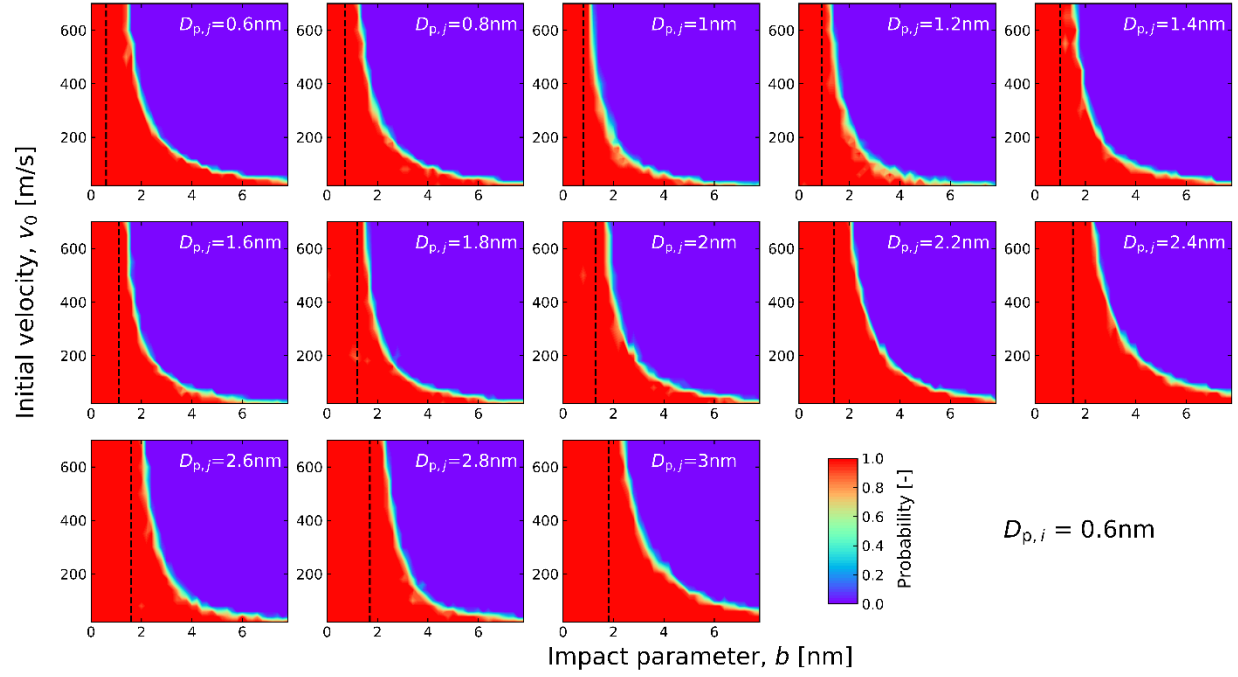


Figure S5. Coagulation probability maps $p(b, v_0)$ with $D_{p,i} = 0.6$ nm and $D_{p,j} = 0.6$ to 3.0 nm. The calculation of $(D_{p,i}, D_{p,j}) = (0.6$ nm, 1.0 nm) was not performed but the result was referred from $(D_{p,i}, D_{p,j}) = (0.6$ nm, 1.0 nm).

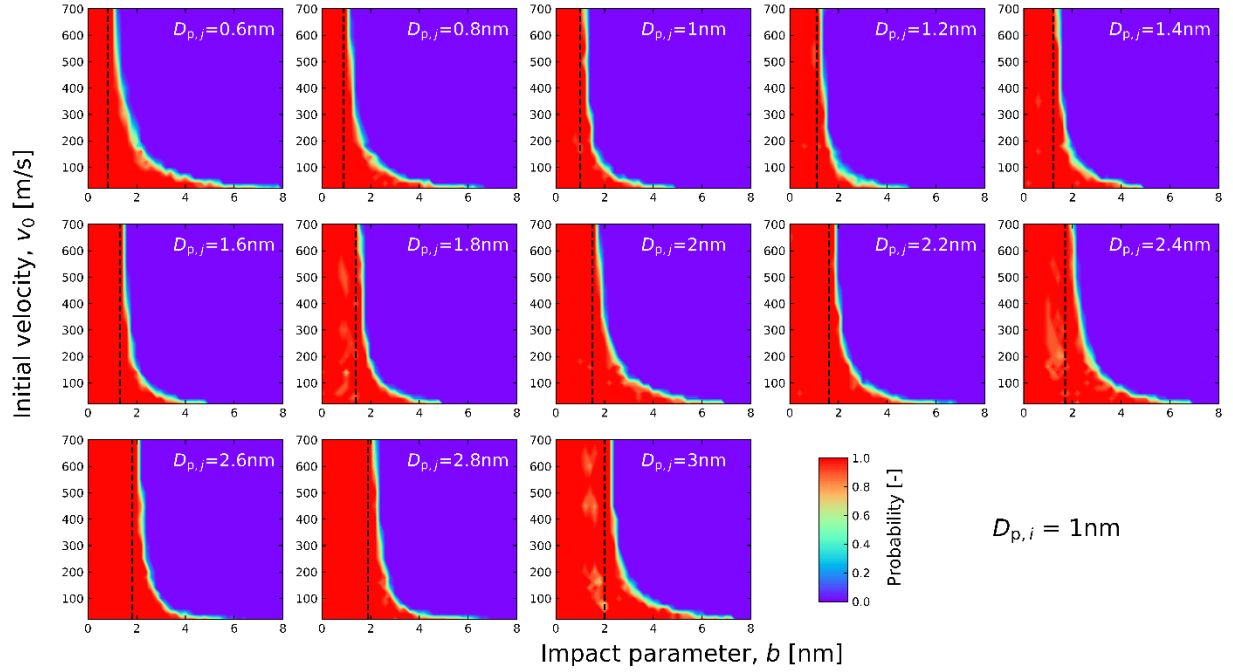


Figure S6. Coagulation probability maps $p(b, v_0)$ with $D_{p,i} = 1.0$ nm and $D_{p,j} = 0.6$ to 3.0 nm.

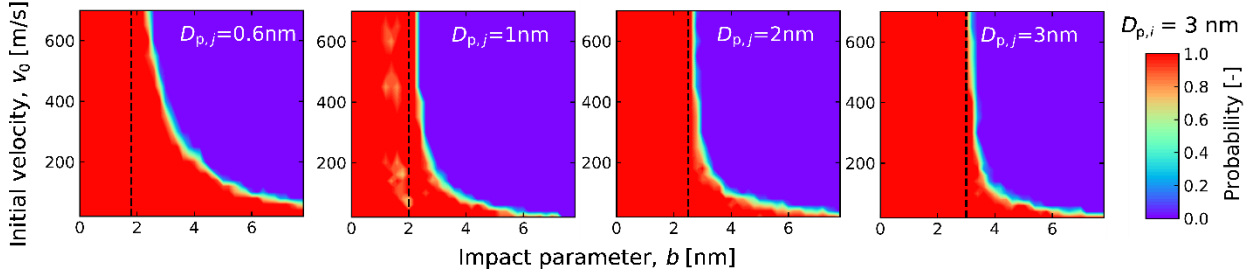


Figure S7. Coagulation probability maps $p(b, v_0)$ for $D_{p,i} = 3.0$ nm and $D_{p,j} = 0.6, 1.0, 2.0,$ and 3.0 nm.

4. Local minimum nanocluster diameter in hard sphere coagulation theory

The hard sphere coagulation rate coefficient is expressed as equation 3 in the main text. In it, a_{ij} and m_{ij} are given as:

$$\frac{1}{m_{ij}} = \left(\frac{1}{m_i} + \frac{1}{m_j} \right)$$

$$a_{ij} = \frac{D_{p,i} + D_{p,j}}{2}$$

m_i and m_j are the masses of two nanoclusters. Approximating the nanoclusters as spheres, these parameters can be expressed as $m_i = \rho \frac{\pi D_{p,i}^3}{6}$ and $m_j = \rho \frac{\pi D_{p,j}^3}{6}$. Substituting the above equations into equation 3 of the main text and differentiating β_{HS} with respect to $D_{p,j}$ yields:

$$\frac{\partial \beta_{\text{HS}}}{\partial D_{p,j}} = 2(D_{p,i} + D_{p,j}) \sqrt{\frac{3k_b T}{\rho} \left(\frac{1}{D_{p,i}^3} + \frac{1}{D_{p,j}^3} \right)} - \frac{3(D_{p,i} + D_{p,j})^2}{2D_{p,i}^4} \sqrt{\frac{3k_b T}{\rho} \left(\frac{1}{D_{p,i}^3} + \frac{1}{D_{p,j}^3} \right)}^{-\frac{1}{2}}$$

The local minimum is observed when $\frac{\partial \beta_{\text{HS}}}{\partial D_{p,j}} = 0$, which leads to:

$$\left(\frac{D_{p,j}}{D_{p,i}} \right)^3 + \frac{1}{4} - \frac{3}{4} \frac{D_{p,i}}{D_{p,j}} = 0$$

The solution of this equation $\frac{D_{p,j}}{D_{p,i}} = 0.856$. This is a local minimum nanocluster diameter ratio in hard sphere coagulation theory for $D_{p,i}$, i.e. the hard-sphere minimum rate for a nanocluster is always with a smaller nanocluster.

5. Losses & accuracies

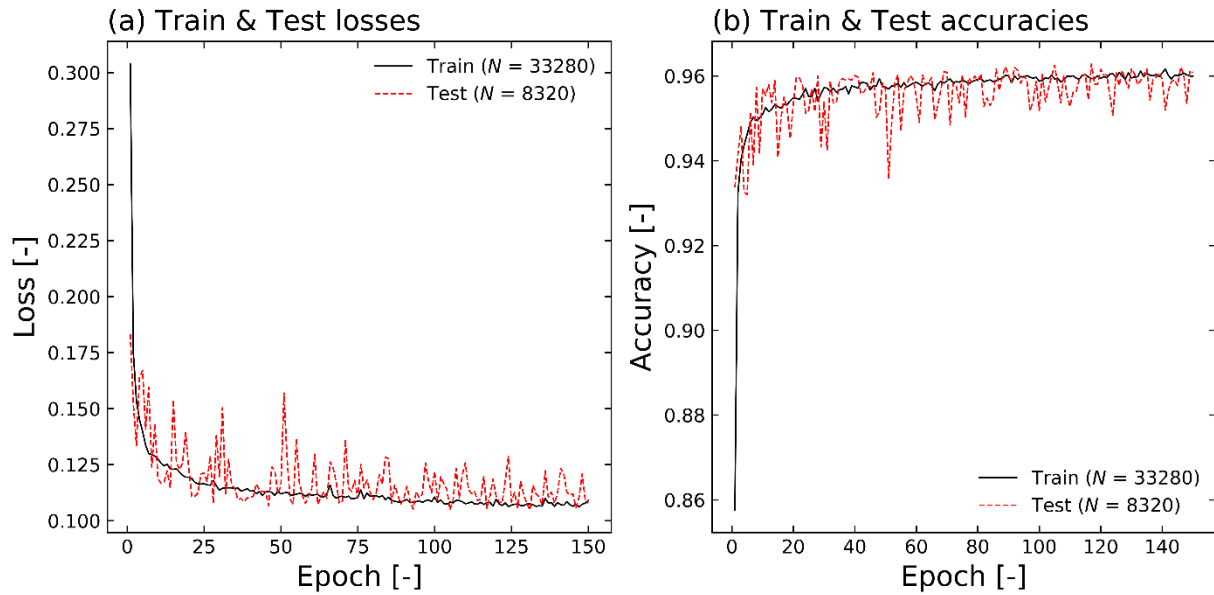


Figure S8. Neural network training and test losses **(a)** and accuracies **(b)** as a function of epoch.

6. Logistic Regression model

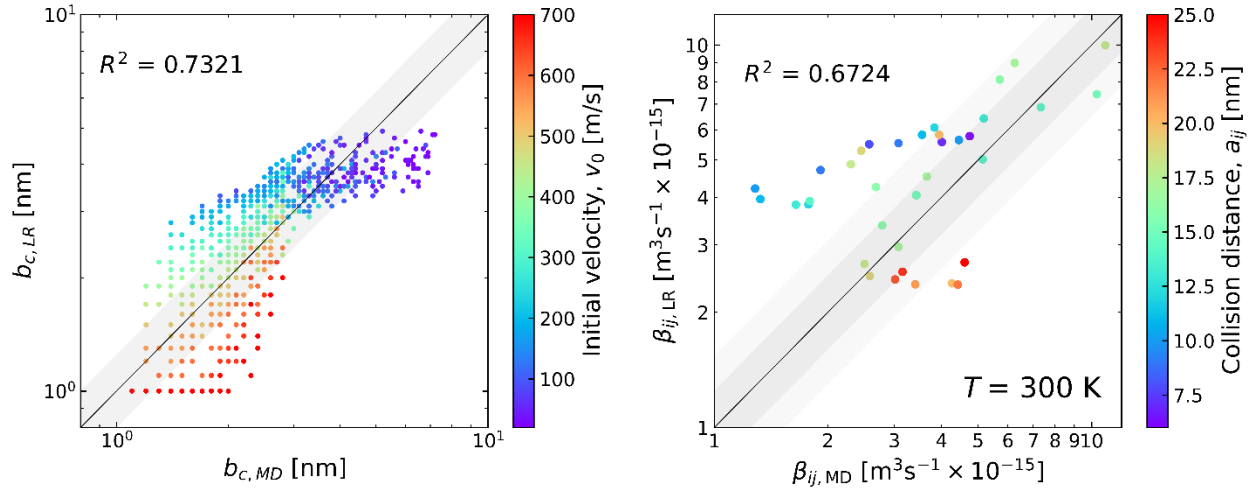


Figure S9. Logistic regression predicted versus MD predicted critical impact parameters, b_c (**left**), coagulation rate coefficient, β_{ij} at 300 K (**right**). The plot color denotes the initial velocity, v_0 in (**left**) and collision distance a_{ij} in (**right**). The diagonal line is the 1:1 line ($b_{c,LR} = b_{c,MD}$ or $\beta_{ij,LR} = \beta_{ij,MD}$), dark gray shadings are 0.8 to 1.25 times from the 1:1 line, and light gray shadings are 0.65 to 1.54. R^2 is the coefficient of determination.